

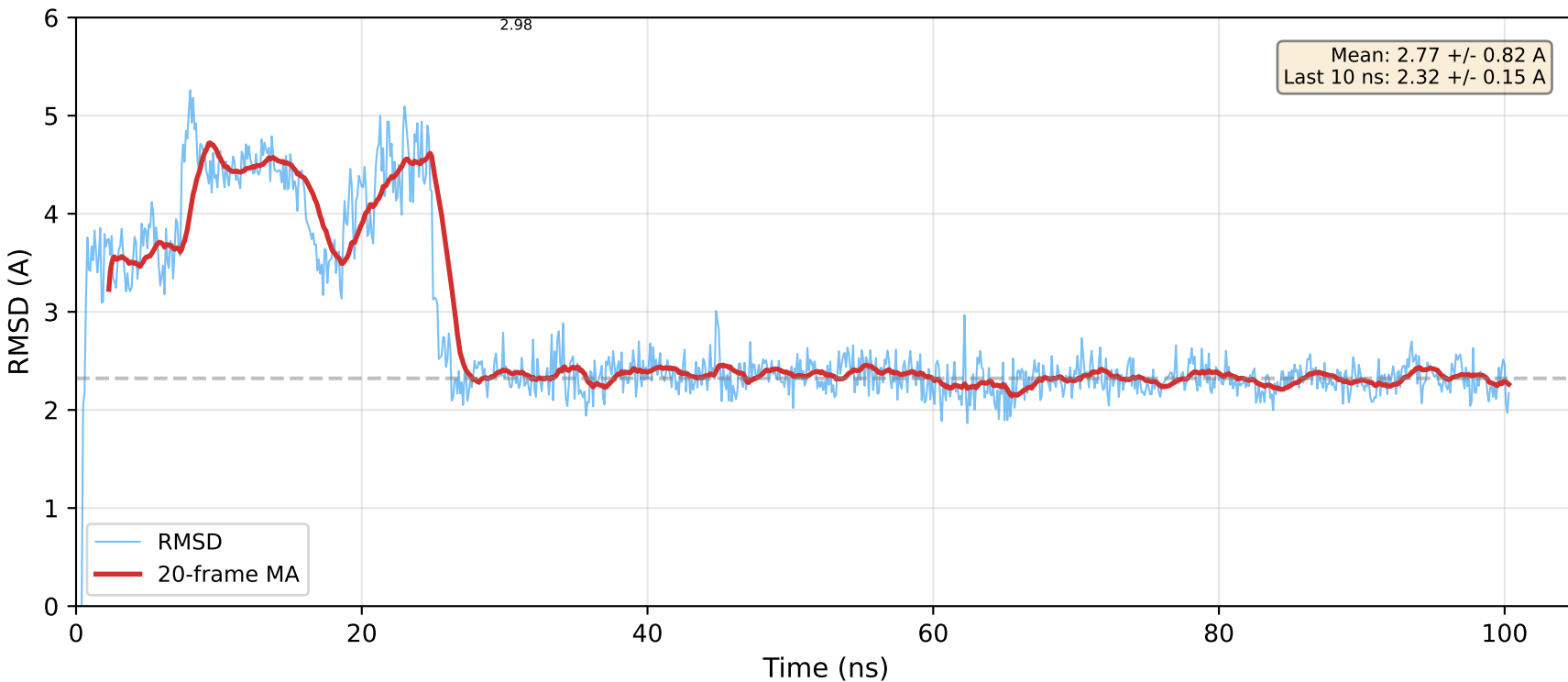
Trp-Cage Miniprotein

100 ns Molecular Dynamics Simulation Report

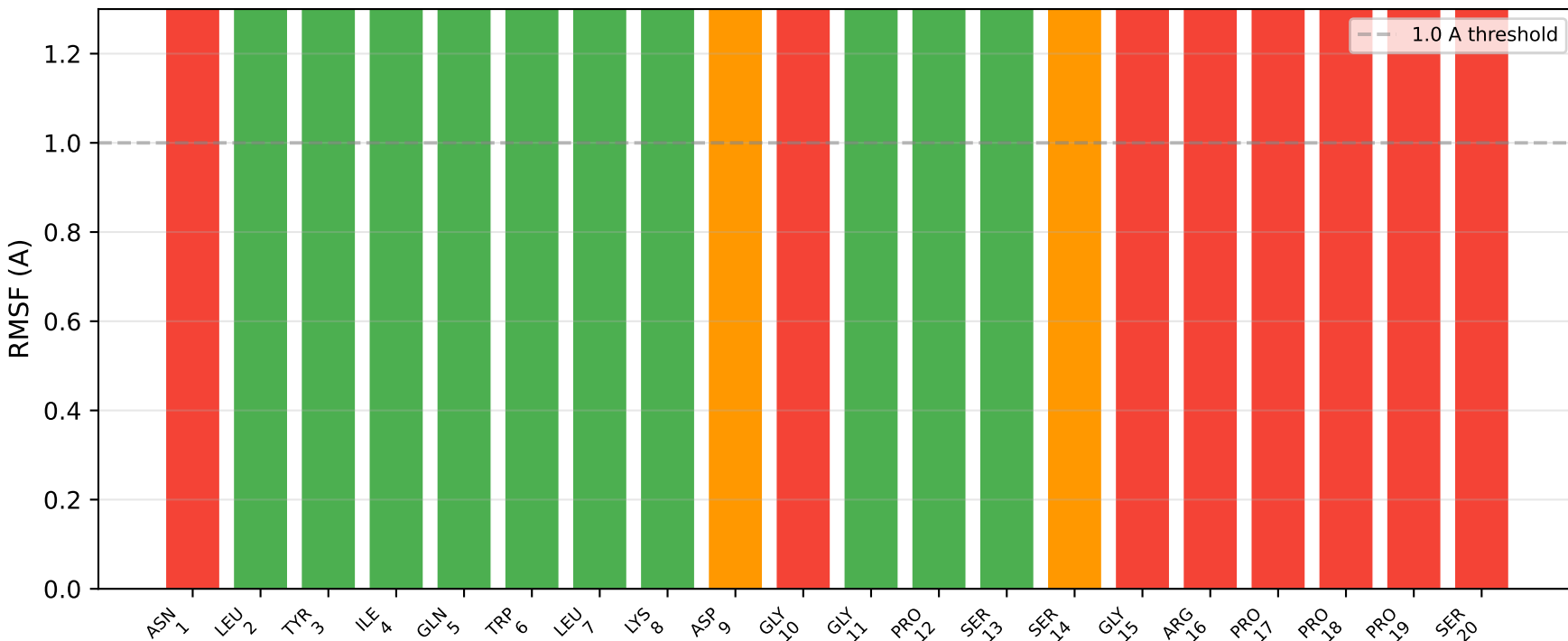
PDB: 1L2Y | Sequence: NLYIQWLKDGG LSSGRPPPS | Force Field: AMBER14 | Solvent: TIP3P

System: 20 residues, 1499 waters, 9 ions | Box: 36.6 Å cubic | Frames: 1000 | dt = 100 ps

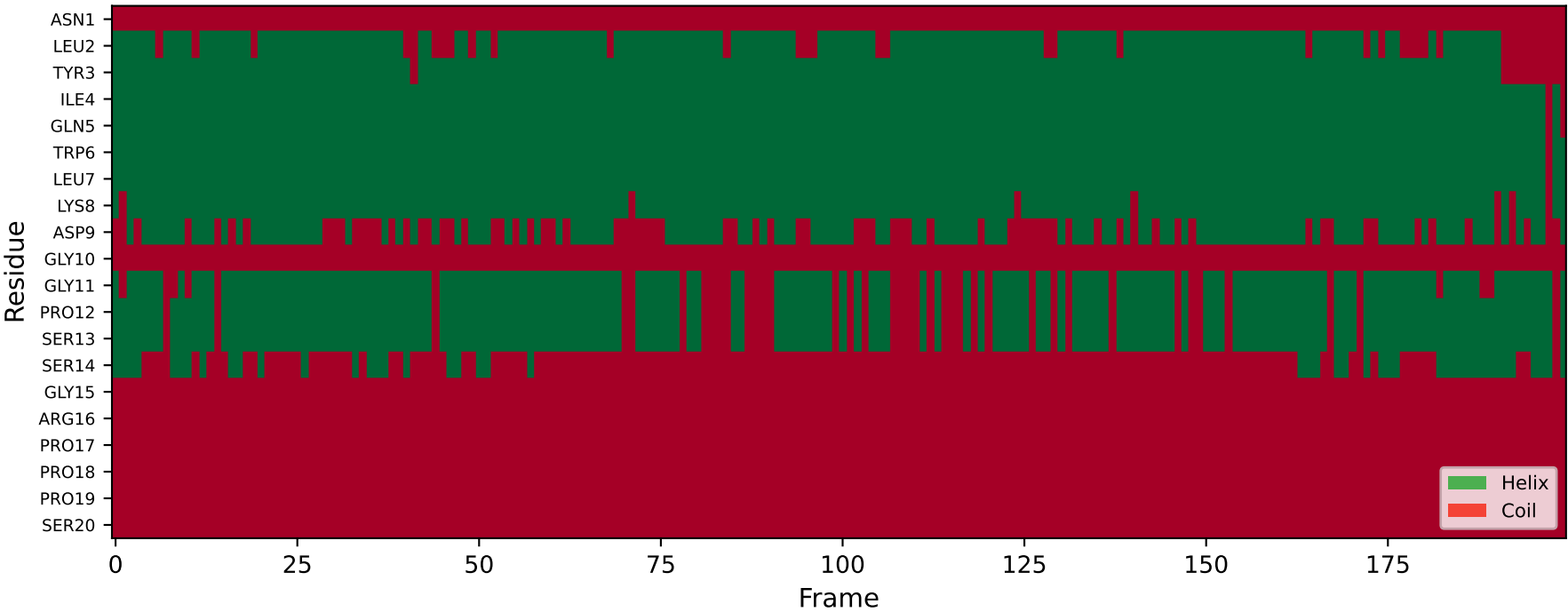
Backbone RMSD to First Frame



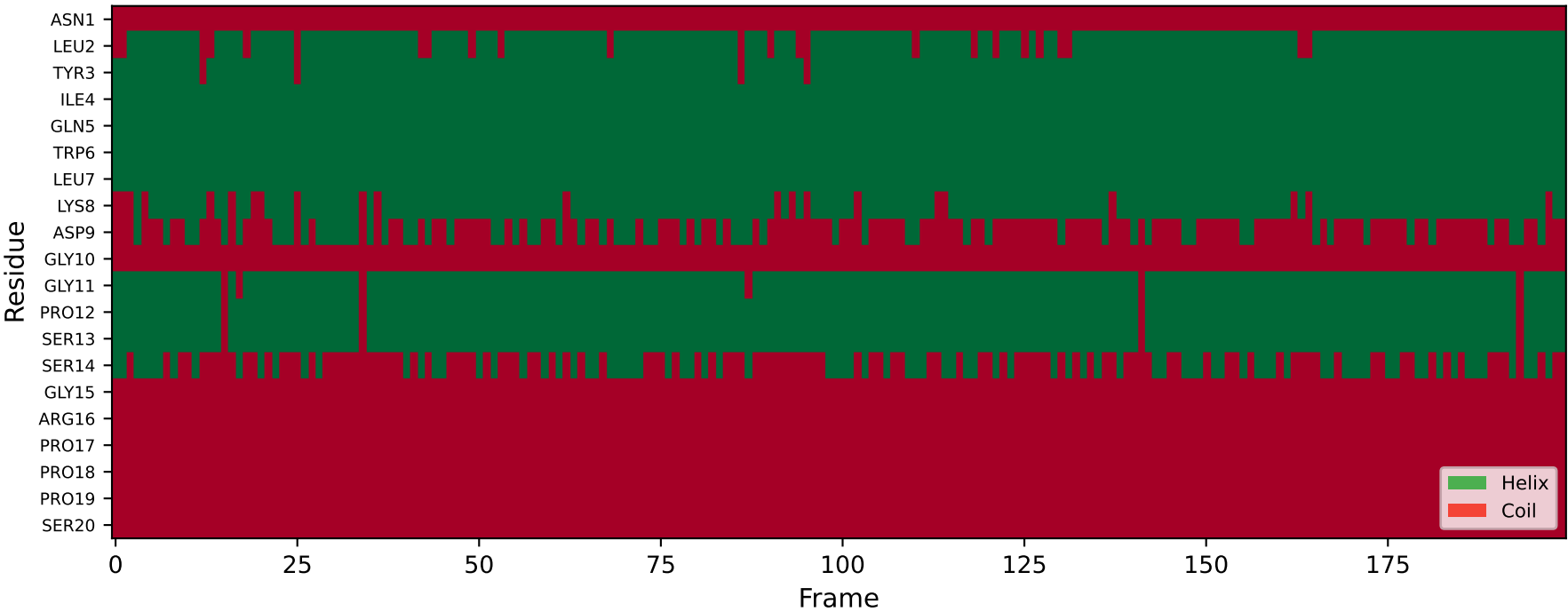
Per-Residue RMSF (color = helix content: green>50%, orange>10%, red<10%)



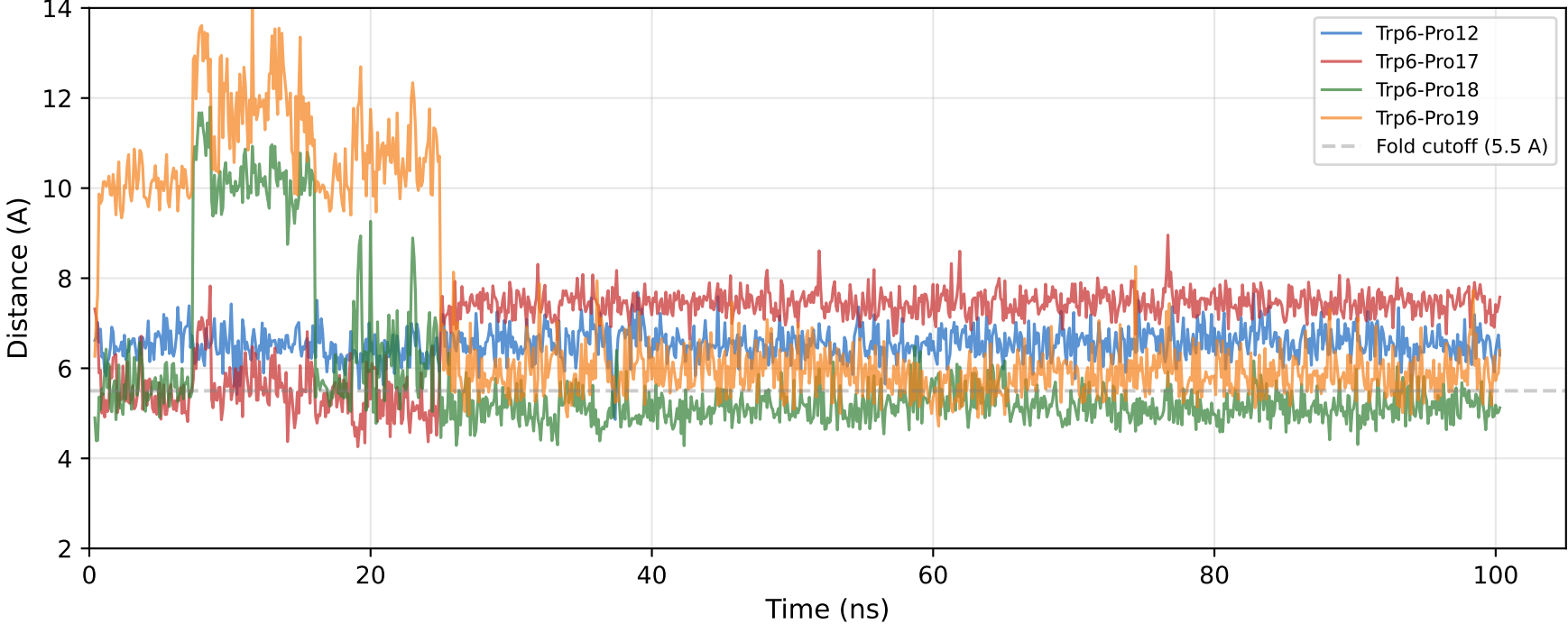
Secondary Structure Evolution (first 20 ns)



Secondary Structure Evolution (last 20 ns)



Trp6-Proline Cage Distances Over Time



Summary Statistics

Metric	Value	Assessment
Simulation Length	100.3 ns (1000 frames)	Complete
RMSD (overall)	2.77 +/- 0.82 Å	Stable
RMSD (last 10 ns)	2.32 +/- 0.15 Å	Converged
RMSF (Trp6 core)	2.98 Å	Very stable (< 0.5 Å)
RMSF (max, Ser20)	10.81 Å	Flexible terminus
Helix content	51.7%	Normal (~50%)
Trp6-Pro12	6.3 Å	Slightly open (cutoff 5.5 Å)
Trp6-Pro18	5.1 Å	Borderline (cutoff 5.0 Å)

Verdict

TRP-CAGE FOLDING ASSESSMENT

PASS Alpha Helix: Stable and persistent (99% for core residues 2-12)
PASS RMSD Convergence: 2.32 Å in last 10 ns (stable < 3 Å threshold)
PASS Core Stability: Trp6 RMSF = 0.30 Å (buried, rigid)
WARN Cage Closure: Trp6-Pro12 = 6.3 Å (target < 5.5 Å)
WARN Cage Closure: Trp6-Pro18 = 5.1 Å (target < 5.0 Å)

OVERALL: The helix is correctly folded and stable. The proline "cage" has not fully wrapped around Trp6 during the 100 ns simulation. This is a known dynamic region -- the cage closure is a slow process that may require longer simulations (200-500 ns) or enhanced sampling methods to observe.

The simulation is physically reasonable: the helix formed and stayed stable, the core (Trp6) is locked in place, and the RMSD converged. The cage dynamics are consistent with the known folding pathway of Trp-cage.